A joint moment bound for the three largest graph eigenvalues

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Abstract

Let $\lambda_1(G) \geq \lambda_2(G) \geq \cdots$ be the adjacency eigenvalues of a finite simple graph. We prove $\lambda_1(G) + \lambda_2(G) + \lambda_3(G) \leq cn - 3$ for every graph of order $n \geq 3$, where $c = 1.284673255723918505\ldots$ is an explicit algebraic constant. This improves the coefficient $(3+\sqrt{5})/4$ in the universal bound of Sivashankar, Tang and Wakhare. The proof controls the signed mean and the absolute Gram energy of a single isotropic law by a joint third- and fourth-order moment identity. Radial and directional errors are retained in the same identity. We also prove that the canonical eigenfunction law of every maximizing graphon has at most 24 support points. The structural argument uses moment-preserving variations and a separating family of binary observations to rule out minimal second-moment dependencies on positive cliques. A small final improvement comes from positive-mass self-pairs. Neither the optimal coefficient nor the extremal graphs are determined.

Keywords. Adjacency eigenvalues; spectral sums; graphons; isotropic measures; projection constants.

1 Introduction

For a finite simple undirected graph G of order $n \geq 3$, put

$$\Lambda_3(G) = \lambda_1(G) + \lambda_2(G) + \lambda_3(G),$$

where the adjacency eigenvalues are ordered algebraically. The signs in this sum are retained. The extremal problem asks how large $\Lambda_3(G)$ can be when only n is prescribed. The spectral-sum problem appears in Ebrahimi's thesis [5]; general bounds for sums of leading eigenvalues were developed by Mohar [11]. More recently, Sivashankar, Tang and Wakhare [12, Proposition 6.1] proved

$$\Lambda_3(G) \leq \frac{3+\sqrt{5}}{4} n - 3. \tag{1.1}$$

Their argument uses the rank-three absolute projection constant $\gamma(3) = (1+\sqrt{5})/2$; a proof of the requisite projection bound is provided by Deręgowska and Lewandowska [3].

The two quantities in that reduction are naturally coupled. If $X \in \mathbb{R}^3$ satisfies $\mathbb{E}XX^T = I_3$ and X' is an independent copy, they are

$$A = \mathbb{E}|X \cdot X'|, \qquad B = |\mathbb{E}X|^2.$$

The separate estimates $A \leq (1+\sqrt{5})/2$ and $B \leq 1$ give (1.1). A better graph bound therefore requires a quantitative restriction on their simultaneous size. The mixed projection constant expressing this question was already introduced in [12, Section 6.2]. We work directly with its isotropic-law formulation.

Define

$$c = \frac{106106902139519}{107310156800000} + \frac{41\sqrt{3}}{240} - \frac{53}{9000000} = 1.284673255723918505\ldots. \tag{1.2}$$

Theorem 1.1. *Every finite simple unweighted graph G on $n \geq 3$ vertices satisfies*

$$\Lambda_3(G) \leq cn - 3. \tag{1.3}$$

Moreover, every Borel probability law on $\mathbb{R}^3$ satisfying $\mathbb{E}XX^T = I_3$ obeys

$$\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2 \leq 2c. \tag{1.4}$$

Only a finite second moment is assumed; zero vectors and unbounded radii are allowed.

The coefficient decrease from (1.1) is $0.024343738651028919\ldots$, with the constant term -3 unchanged. This comparison concerns the stated universal order-only bound. It does not compare all estimates involving additional graph parameters, nor assert that c is optimal.

We use graphons to pass to an attained maximum. For a symmetric measurable kernel $W : [0,1]^2 \to [0,1]$, let

$$\Sigma_k(W) = \sup_{\mathbb{E}FF^T = I_k} \iint W(x,y)F(x) \cdot F(y)\, dx\, dy, \qquad \mathcal{M} = \sup_W \Sigma_3(W). \tag{1.5}$$

The supremum is over real orthonormal k-frames. Since $W \in L^2([0,1]^2)$, the operator T_W is Hilbert–Schmidt and therefore compact; it is also self-adjoint, so its spectrum is real and accumulates only at zero, and Σ_k is the sum of the first k positive eigenvalues, padded by zero if there are fewer than k of them. This convention applies only to graphons, not to $\Lambda_3(G)$. The maximum $\mathcal{M}$ is attained. For a maximizing graphon, choose an orthonormal top-three eigenframe $F = (f_1, f_2, f_3)^T$, with $f_1 \geq 0$, and call $F_{\#}dx$ its canonical law.

Theorem 1.2. *Let W attain $\mathcal{M}$, let $\sum_i p_i \delta_{v_i}$ ($p_i > 0$) be its canonical law, and let L be the diagonal matrix of the three leading eigenvalues of W. Then:*

(a) the support has at most 24 points;

(b) W agrees almost everywhere with the binary step kernel $\mathbf{1}_{\{v_i \cdot v_j > 0\}}$ on the corresponding positive-measure fibers;

(c) the support points satisfy

$$\sum_i p_i v_i v_i^T = I_3, \qquad v_i^T L v_i = \mathcal{M}, \qquad v_{i1} > 0,$$

$$v_i \cdot v_j \neq 0, \qquad B_{ij} = \mathbf{1}_{\{v_i \cdot v_j > 0\}}, \qquad \sum_j p_j B_{ij} v_j = L v_i;$$

(d) for every positive-dot clique J in the support, the matrices $\{v_i v_i^T : i \in J\}$ are linearly independent; in particular $|J| \leq 6$.

This assertion concerns every original maximizer, not merely the existence of a maximizer with small support.

1.1 Method and relation to earlier work

The proof has two parts. First, second variation on the actual probability space gives a negative semidefinite kernel. On each of four sign classes, this controls the rank of the positive-part Gram kernel and forces finite canonical support. Moment-preserving variations then imply vector relations against every binary row of the maximizing kernel. Those rows separate distinct support points. A minimal-dependency argument shows that no second-moment relation can survive on a positive clique.

The variation, cloning, and orthant-reduction mechanisms have close antecedents in Kumar, Mohar, Mojallal and Pragada [8]. Related graphon ellipse and clique-cloning arguments for the two-eigenvalue problem appear in Kumar, Liu, Monterde, Pragada and Tait [7]. General bounds for minimal dependencies of powers of linear forms are due to Li and Liu [9]. Our separating-observation lemma is stated and proved with its specific hypotheses in Section 4; we do not claim a new general bound for such dependencies.

Second, for one isotropic law we retain the mean deficit, radial variance, regression error, and weighted harmonic moments of degrees three and four. A signed quartic majorant couples them in one exact identity. Directional Hermite inequalities and a radial comparison lead to an explicit sum of nonnegative terms. Harmonic moment and polynomial-majorant methods are classical; see Glazyrin [6], Deręgowska and Lewandowska [3, 4], and Sivashankar, Tang and Wakhare [12]. The contribution here is the particular joint control of these errors for the mixed signed and absolute energy, rather than the use of polynomial majorants by itself. The support theorem supplies a small additional term through self-pairs of actual atoms. Diagonal removal and effective-cardinality estimates are standard weighted-energy devices; see [1].

Sections 2–4 prove the structural result. Sections 5–8 give the joint moment certificate. Section 9 applies it to a genuine maximizer and transfers the bound to all isotropic laws and finite graphs.

2 Variational structure of a maximizing graphon

We may work on any standard nonatomic probability space. All changes of measure below can be realized on such a space.

Lemma 2.1. *Every* $[0,1]$*-valued graphon satisfies* $\Sigma_2(W) \le 7/6$.

Proof. We give the elementary argument for the rank-two absolute-energy bound needed below. Take any $\mathbb{R}^2$-valued law $\mathbb{E}YY^T = I_2$, write $Y = RU$ with a unit choice at zero, $C = \mathbb{E}R$, and $N = \mathbb{E}[RUU^T]$. Then $C^2 \le 2$ and $\operatorname{tr}(N^2) \le 1$, the latter from $\mathbb{E}\|U - NY\|^2 = 1 - \operatorname{tr}(N^2) \ge 0$. On the circle, $g_{4,2}(t) = t^4 - t^2 + 1/8$ is a positive kernel: if $U = (\cos a, \sin a)$, it is $\cos(4(a-a'))/8$. Hence its RR'-weighted integral is nonnegative. The exact majorant

$$\frac{2}{9} + \frac{11}{9}t^2 - \frac{4}{9}t^4 - |t| = \frac{(1-|t|)(2|t|-1)^2(|t|+2)}{9} \ge 0$$

therefore gives

$$\mathbb{E}|Y \cdot Y'| \le \frac{5}{18}C^2 + \frac{7}{9}\operatorname{tr}(N^2) \le \frac{4}{3}.$$

Also $|\mathbb{E}Y|^2 \le 1$. For any orthonormal two-frame in a graphon, $W(Y \cdot Y') \le (Y \cdot Y')_+$ pointwise, so its trace is at most $(4/3 + 1)/2 = 7/6$. Taking the supremum proves the lemma. The absolute-energy estimate is classical; see [6, Theorems 2 and 4]. □

Lemma 2.2. *The supremum* $\mathcal{M}$ *is attained, and*

$$c_{\mathrm{w}} := \frac{9 + \sqrt{5} + 2\sqrt{35}}{18} \le \mathcal{M} \le \frac{1 + \sqrt{3}}{2}.$$

Every maximizing graphon has three positive leading eigenvalues.

Proof. For an orthonormal k-frame, $K(x,y) = F(x) \cdot F(y)$ satisfies $\int K = |\int F|^2 \le 1$ and $\int K^2 = k$. Hence $\int WK \le \int K_+ \le (1 + \sqrt{k})/2$. For the lower bound, use a six-part kernel

with template $K_1 \vee C_5$ and diagonal blocks equal to one. Give the center mass $a = 1 - 5b$ and each rim part mass b, where $0 < b < 1/5$. Put $\phi = (1 + \sqrt{5})/2$. Its quotient spectrum is

$$\frac{1 - 2b \pm \sqrt{1 + 4b - 36b^2}}{2}, \quad \phi b,\ \phi b,\ -b/\phi,\ -b/\phi.$$

Indeed the four nonconstant Fourier modes on the pentagon give the last four values; the remaining block is $\left(\begin{smallmatrix} a & \sqrt{5ab} \\ \sqrt{5ab} & 3b \end{smallmatrix}\right)$. Its positive eigenvalue exceeds $3b$, and its determinant is $-2ab < 0$. Consequently the sum of its three positive eigenvalues is

$$\frac{1 + 2\sqrt{5}b + \sqrt{1 + 4b - 36b^2}}{2}.$$

Substitution of $b = (7 + 5\sqrt{7})/126$ gives c_{w}. This is the usual quotient calculation for a join construction; see [2]. In particular $c_{\mathrm{w}} > 32/25 > 7/6$.

For completeness we record why Σ_3 of this step kernel is exactly the sum of the three positive quotient eigenvalues. If f is supported in one part A_i and $\int_{A_i} f = 0$, then for $x \in A_i$ one has $(T_W f)(x) = \int_{A_i} f = 0$, while for x in any other part A_j one has $(T_W f)(x) = W_{ji} \int_{A_i} f = 0$; hence $T_W f = 0$. The whole block-mean-zero subspace therefore lies in the zero eigenspace, and Σ_3 is governed by the quotient matrix alone.

For attainment, let $D = W_1 - W_2$. If $\lambda \neq 0$ is an eigenvalue of T_D with unit eigenfunction f, then $|D| \leq 1$ gives $\|f\|_\infty \leq 1/|\lambda|$. The cut-norm dual estimate yields

$$|\lambda| = |\langle f, T_D f \rangle| \leq 4\|D\|_\square \|f\|_\infty^2, \qquad \|T_D\|_{2\to 2}^3 \leq 4\|D\|_\square.$$

The assertion is trivial when $T_D = 0$; otherwise compactness supplies an eigenvalue of modulus $\|T_D\|$. By the variational definition, $|\Sigma_3(W_1) - \Sigma_3(W_2)| \leq 3\|T_D\|$. This also holds after relabeling. Cut-metric compactness [10] therefore gives a maximizer. Finally Lemma 2.1 implies $\lambda_3 \geq c_{\mathrm{w}} - 7/6 > 0$ at each maximizer. □

Fix a maximizing W and a top-three orthonormal eigenframe F, taking $f_1 \geq 0$. Write $L = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$. Taking the absolute value of a top Rayleigh maximizer supplies the nonnegative choice of f_1. We do not initially assume any eigenvalue simplicity.

Lemma 2.3. *Almost everywhere, $F^T L F = \mathcal{M}$. Moreover $\lambda_3 > \lambda_4$, where the fourth positive eigenvalue is zero if absent. The canonical support $S = \mathrm{supp}(F_\# dx)$ is compact, lies on the ellipsoid $v^T L v = \mathcal{M}$, and avoids the origin.*

Proof. For a bounded real g of mean zero, change density to $1 + tg$ for both signs of sufficiently small t. Set $Q = \int g F F^T$. The unchanged frame has complete Gram matrix $I + tQ$ and numerator matrix $L + t(QL + LQ) + O(t^2)$. After a common orthonormalization its trace is

$$\mathrm{tr}\big[(I + tQ)^{-1}\{L + t(QL + LQ) + O(t^2)\}\big] = \mathcal{M} + t\,\mathrm{tr}(LQ) + O(t^2).$$

Maximality gives $\int g F^T L F = 0$. Since every bounded mean-zero g is allowed, $F^T L F$ equals its mean $\mathcal{M}$ almost everywhere. This applies to every active eigenframe. If $\lambda_3 = \lambda_4 = \lambda > 0$, compare the frames with third entries f_3, f_4 and $(f_3 + f_4)/\sqrt{2}$, keeping f_1, f_2 fixed. The three identities imply $f_3^2 = f_4^2$ and $f_3 f_4 = 0$ almost everywhere, a contradiction. Only finitely many null sets are combined. Since $L \succ 0$, the ellipsoid is compact and excludes zero. It contains the full support, not just almost every frame value. □

Lemma 2.4. *For every measurable $h : [0,1] \to [0,1]$, the vector $r = \int hF$ satisfies $r^T L^{-1} r \leq \mathcal{M}$.*

Proof. Give the original space A mass $u = 1 - \epsilon$ and a new independent part B mass ϵ; the kernel is W on $A \times A$, equals the prescribed cross-row h (a function on A) on $A \times B$ and $B \times A$, and is zero on $B \times B$. Put $t = L^{-1}r$ and $T = |t|^2$. For the trial frame take f_i on A and the constant t_i on B; then the $A \times B$ and $B \times A$ contributions to the numerator are $\epsilon u\, r_i t_j$ and its transpose. Dividing the entire frame by $\sqrt{u}$ leaves every Rayleigh quotient unchanged, and the resulting Gram and numerator matrices are

$$G = I + (\epsilon/u)tt^T, \qquad N = uL + \epsilon(rt^T + tr^T).$$

The rank-one inverse formula gives the exact trial trace

$$\operatorname{tr}(G^{-1}N) = u\mathcal{M} + \frac{\epsilon r^T L^{-1} r}{1 + \epsilon T/u}.$$

This cannot exceed $\mathcal{M}$. Divide by ϵ and let $\epsilon \downarrow 0$. □

Lemma 2.5. *For $\nu = F_{\#}dx$ and every $v \in S$,*

$$\nu\{w : v \cdot w = 0\} = 0, \qquad \int_{v \cdot w > 0} w \, d\nu(w) = Lv, \qquad v_1 > 0. \tag{2.1}$$

The original kernel is $\mathbf{1}_{\{F(x)\cdot F(y)>0\}}$ almost everywhere. It is irreducible, and $\lambda_1 > \lambda_2$.

Proof. Let $Z = \{\int hF : 0 \le h \le 1\}$ and $E = \{r : r^T L^{-1} r \le \mathcal{M}\}$. The set Z is compact by weak-star compactness and is contained in E by Lemma 2.4. Its support function is $H(v) = \int (v \cdot F)_+$. For almost every x, the original row has integral vector $LF(x)$, and $F(x)^T LF(x) = \mathcal{M}$. Since the support function of E is $\sqrt{\mathcal{M} v^T L v}$, it follows that $H(F(x)) = \mathcal{M}$. Continuity then gives $H(v) = \mathcal{M}$ for every $v \in S$.

For such v, the unique support point of the strictly convex ellipsoid E in direction v is Lv. The row $h_v = \mathbf{1}_{\{v\cdot F>0\}}$ attains this support value, so $\int h_v F = Lv$. If $A_0 = \{v \cdot F = 0\}$, then $h_v + \mathbf{1}_A$ has the same support value for every measurable $A \subset A_0$. Uniqueness forces $\int_A F = 0$ for every such A, hence $F = 0$ almost everywhere on A_0. The ellipsoid identity implies that A_0 is null.

It remains to identify every maximizing row, not merely those obtained by perturbing h_v on A_0. Let $h \in [0, 1]$ satisfy $\int hF = Lv$ for some $v \in S$. Pairing with v and using $v^T Lv = \mathcal{M}$ gives $v \cdot \int hF = \mathcal{M}$, while on the other hand

$$v \cdot \int hF = \int h\,(v \cdot F)\, dx \le \int (v \cdot F)_+ \, dx = H(v) = \mathcal{M}.$$

Hence equality holds throughout. Since $0 \le h \le 1$, the pointwise bound $h\,(v \cdot F) \le (v \cdot F)_+$ is therefore an equality almost everywhere, which forces $h = 1$ almost everywhere on $\{v \cdot F > 0\}$ and $h = 0$ almost everywhere on $\{v \cdot F < 0\}$. As A_0 is null, $h = h_v$ almost everywhere. Thus the maximizing row is unique modulo null sets, which identifies every original row and proves the threshold formula.

If a nontrivial measurable cut had zero cross-kernel, its indicator multiplier would commute with T_W and with the isolated top-three spectral projection. The kernel $F(x) \cdot F(y)$ of that projection would vanish across the cut, contradicting the zero-equator assertion and Fubini. Thus W is irreducible. The zero set of the nonnegative f_1 would also give a zero cut by its eigen-equation, so $f_1 > 0$ almost everywhere. If another top eigenfunction orthogonal to f_1 existed, it would change sign. Equality in the absolute-value Rayleigh comparison would force zero edges between its positive and negative sets; its absolute value is a nonnegative top eigenfunction and is positive almost everywhere. This contradicts irreducibility, proving simplicity. Finally, for every $v \in S$, $\lambda_1 v_1 = \int h_v f_1 > 0$: otherwise $h_v = 0$ almost everywhere and hence $Lv = 0$, which is impossible. □

3 Finite canonical support

Continue with the maximizing graphon and canonical law of the previous section, and write $h_v(w) = \mathbf{1}_{\{v\cdot w>0\}}$. Define continuous symmetric kernels on $S \times S$ by

$$A(v,w) = (v\cdot w)_+, \qquad D(v,w) = (v\cdot w)(v^T L w), \qquad K = A - D.$$

Theorem 3.1 (Second variation on the actual support)**.** *For every finite list $v_1,\dots,v_s \in S$ and every real list $a_1,\dots,a_s$,*

$$\sum_{i,j} a_i a_j K(v_i,v_j) \le 0. \tag{3.1}$$

The assertion concerns only the actual canonical support.

Proof. For any bounded measurable real g on S with $\int g\,d\nu = 0$, use the legitimate density $1 + tg(F(x))$ for both signs of sufficiently small t. Set

$$Q = \int g(v) v v^T\, d\nu(v), \qquad R = \iint h_v(w) v w^T g(v) g(w)\, d\nu(v) d\nu(w).$$

The unchanged three coordinate functions have the complete Gram $G_t = I + tQ$ and numerator matrix $N_t = L + t(QL + LQ) + t^2 R$ in this changed measure. For small t, G_t is positive definite; multiplying the frame by $G_t^{-1/2}$ is a genuine common orthonormalization. Since $N_t = G_t L G_t + t^2(R - QLQ)$ and $\operatorname{tr}(LQ) = \mathcal{M}\int g = 0$, its exact trace is

$$\operatorname{tr}(G_t^{-1} N_t) = \mathcal{M} + t^2 \operatorname{tr}(G_t^{-1}(R - QLQ)) \le \mathcal{M}. \tag{3.2}$$

Letting $t \to 0$ gives $\iint K(v,w) g(v) g(w)\, d\nu(v) d\nu(w) \le 0$. There is no discarded remainder or varying-scheme constant in this calculation. By Lemma 2.3 and (2.1), $\int K(v,w) d\nu(w) = \mathcal{M} - \mathcal{M} = 0$ for every $v \in S$. Subtracting the mean therefore proves the same inequality for every bounded g. To obtain (3.1), combine repeated points and take disjoint shrinking neighborhoods of the remaining points. Each has positive ν-mass by full support. Use $g = \sum_i a_i 1_{U_i}/\nu(U_i)$, separately for each neighborhood size. This auxiliary g is bounded but need not have mean zero. The legal probability density direction is $g - \int g\,d\nu$; centering leaves the K quadratic form unchanged. Its bound need not be uniform across neighborhood sizes. Continuity of K on compact $S \times S$ gives (3.1) in the limit. This limiting step needs no uniform permissible size of t. No eigenvalue differentiation is used, including when $\lambda_2 = \lambda_3$. □

3.1 Propagation of moment relations

Lemma 3.2. *If $v_i \in S$ have $v_i \cdot v_j > 0$ for all i, j, then*

$$\sum_i a_i v_i v_i^T = 0 \quad\Longrightarrow\quad \sum_i a_i v_i = 0 \quad \textit{and} \quad \sum_i a_i A(v_i, w) = 0 \quad \textit{for every } w \in S.$$

Consequently the vector space spanned by the functions $A(v,\cdot)$, $v \in S$, has dimension $r \le 24$ and admits a basis $A(z_j,\cdot)$ with actual $z_j \in S$.

Proof. Write $Q_a = \sum a_i v_i v_i^T$ and $b_a = \sum a_i v_i$. The quadratic form in (3.1) is $|b_a|^2 - \operatorname{tr}(LQ_a^2)$ on this positive-dot set. For $Q_a = 0$ it is nonnegative, hence zero and $b_a = 0$. A null vector of a negative semidefinite quadratic form has zero cross terms: apply (3.1) also to this list with w appended and coefficient $s \in \mathbb{R}$. The resulting polynomial $2s\sum_i a_i K(v_i,w) + s^2 K(w,w)$ is nonpositive for all s, so its linear coefficient vanishes. The D sum vanishes because $Q_a = 0$; therefore the A sum vanishes for every w. Partition S by the signs of coordinates 2 and 3, assigning zero to the positive sign. Within each of the four disjoint sets, every dot product

is strictly positive, since coordinate 1 is positive and the other two coordinate products are nonnegative. In each nonempty set choose a basis, of size at most 6, of its matrices vv^T in the six-dimensional space of symmetric 3 by 3 matrices. The implication just proved expresses every corresponding A row in the span of those basis-point rows. Combine the four lists and discard dependent rows. This proves $r \le 24$. These are spanning witnesses, not a replacement vertex measure. This bounds the row space, not yet the cardinality of the support. □

3.2 Sign regions and finite support

Lemma 3.3 (Sign-region dimension descent)**.** *Let $T \subset S$ have linear span of dimension $d \ge 1$. Suppose for every $w \in S$ there is a vector $b_w \in \mathbb{R}^3$ such that $A(v, w) = v \cdot b_w$ for all $v \in T$. Then $|T| \le 2^{d-1}$.*

Proof. If $d = 1$, the positive first coordinate and $v^T L v = \mathcal{M}$ make T a singleton. For general d, if no $w \in S$ has strictly positive dot product with one member of T and strictly negative dot product with another, any two members have the same threshold row outside the union of their two zero equators. Equation (2.1) then makes the members equal. Otherwise choose such a w, with witnesses v_+, v_-, and split $T_+ = \{v \in T : v \cdot w \ge 0\}$ and $T_- = \{v \in T : v \cdot w < 0\}$. On T_+, $v \cdot (b_w - w) = 0$; on T_-, $v \cdot b_w = 0$. Both are proper hyperplanes of $\operatorname{span} T$: evaluate $b_w - w$ at v_- and b_w at v_+. Thus both children have span dimension at most $d - 1$. They inherit the linearization for every $w' \in S$. Induction gives at most 2^{d-2} points in each child, proving the bound. Zeros were included in T_+, so none are lost in the dimension descent. □

Theorem 3.4 (Finite support of every maximizer)**.** *For every global maximizer of Σ_3, its actual canonical support satisfies*

$$|S| \le 4(1 + r(r+1)/2) \le 1204 =: N_*. \tag{3.3}$$

The original graphon agrees almost everywhere with a binary step graphon on these actual fibers, with positive masses, diagonal blocks 1, and no equal rows. For this particular maximizer $\min_{v,w \in S} |v \cdot w| > 0$.

Proof. Take the actual anchors $z_1, \dots, z_r$ from Lemma 3.2. Symmetry gives, for each $w \in S$, coefficients $c_j(w)$ with $A(v, w) = \sum_j c_j(w) A(v, z_j)$ for every $v \in S$. The union of the finitely many anchor equators has ν-measure zero by (2.1). On any fixed strict sign region of these anchors, $A(v, z_j)$ is either $v \cdot z_j$ or zero. Thus $A(v, w)$ is linear in v there, for every w, with coefficients determined by that region. Because $v_1 > 0$, use coordinates $(v_2/v_1, v_3/v_1)$. Each anchor equator becomes an affine line, or is empty if its last two coordinates vanish. At most r lines have at most $1 + r(r+1)/2$ complementary regions: the j-th line has at most $j - 1$ distinct previous intersection points and adds at most j regions. Coincident or parallel lines only lower this count. A strict sign pattern is an intersection of open half-planes, hence connected whenever nonempty. Apply Lemma 3.3 to $T = S$ intersected with each such sign region; its span dimension is at most 3, so it contains at most four points. We have therefore placed full ν-measure in a finite set of size bounded by (3.3). Its closure is the same finite set, so full support of ν implies that all of S is contained in it, including any previously excluded boundary candidates. Every support point is now an atom of positive mass. The individual zero-equator statements in (2.1) imply $v \cdot w \ne 0$ for every pair of support atoms. Taking a minimum over this finite set proves the asserted margin. The actual fibers $F^{-1}(v)$ provide the partition, and (2.1) identifies its rows. □

The margin asserted here may depend on the maximizer.

4 Separating binary observations

By the preceding section, write $S = \{v_1, \dots, v_m\}$ with actual masses $p_i > 0$. Put $V = (v_i^T)_{i=1}^m$, $P_0 = \operatorname{diag}(p_i)$ and $B_{ij} = \mathbf{1}_{\{v_i \cdot v_j > 0\}}$. Then

$$BP_0V = VL, \qquad \sum_i p_i v_i v_i^T = I_3, \qquad v_i^T L v_i = \mathcal{M}. \tag{4.1}$$

The full rows of B are distinct: equal rows would imply equal v_i by invertibility of L. A positive clique J means a subset of this actual support with $B_{ij} = 1$ for every $i, j \in J$. Extend vectors indexed by J by zero outside J.

Lemma 4.1 (Vector rigidity on a positive clique)**.** *For every real a supported on a positive clique, if $Q_a = \sum_i a_i v_i v_i^T = 0$, then*

$$\sum_i a_i = 0, \quad b_a := \sum_i a_i v_i = 0, \quad \sum_i B_{ki} a_i v_i = 0 \quad \textit{for every row } k. \tag{4.2}$$

Proof. Taking the trace against L gives $\mathcal{M} \sum a_i = 0$. For both signs of sufficiently small nonzero t, all $p_i + t a_i$ are positive. These are legitimate new fiber measures for the same binary template, realizable on an atomless space. They are not independent choices of moment data. The original coordinate functions have Gram matrix $I + tQ_a = I$ and compression

$$N(t) = L + t(Q_a L + L Q_a) + t^2 b_a b_a^T = L + t^2 b_a b_a^T.$$

The last term uses the clique assumption, and both cross terms use (4.1). Their trace cannot exceed the global maximum $\mathcal{M}$; hence $b_a = 0$ and $N(t) = L$. Equality of this trial trace with the maximum forces its three-dimensional space to be invariant. Explicitly, rotate one orthonormal coordinate into any unit perpendicular function; the first derivative of the trace gives zero off-space matrix element. This argument does not differentiate ordered eigenvalues. The compression L then gives the eigenfunction equations at the perturbed measure. All fibers still have positive mass, so for every k, $\sum_i (p_i + t a_i) B_{ki} v_i = L v_k$. Subtract (4.1) and divide by t to obtain (4.2). □

Moment-preserving cloning and the Ky Fan equality argument follow the mechanism in [8, Claims 2.2–2.5]. The permissible size of t may depend on the maximizer and the chosen direction.

4.1 A separating observation lemma

Minimal dependencies of rank-one powers have a broader theory; see Li and Liu [9]. The following lemma uses the additional binary separation hypothesis. Its elementary proof is included.

Lemma 4.2 (Moment independence from separating binary observations)**.** *Let $T = \{v_i : i \in J\}$ be a finite set of distinct points in $\mathbb{R}^3$ with $v_{i1} > 0$ and $v_i^T L v_i = \mathcal{M} > 0$, where L is positive definite. Let $\mathcal{B}$ be a family of functions $b : J \to \{0, 1\}$ containing the constant-one function and separating every pair of distinct indices. Suppose, for every real a with $\sum_{i \in J} a_i v_i v_i^T = 0$, that*

$$\sum_{i \in J} a_i b(i) v_i = 0 \quad \textit{for every } b \in \mathcal{B}. \tag{4.3}$$

Then the matrices $\{v_i v_i^T : i \in J\}$ are linearly independent.

Proof. Suppose instead that $I \subseteq J$ is a support-minimal dependent family of these matrices, with relation $a_i \neq 0$ for all $i \in I$. Write $s = |I|$. Since $\dim \mathrm{Sym}_3 = 6$, minimality gives $s \leq 7$. The ellipse equation gives $\sum_I a_i = 0$, while the constant observation in (4.3) gives $\sum_I a_i v_i = 0$. With $w_i = (1, v_i)^T \in \mathbb{R}^4$, all three identities combine to give

$$\sum_{a_i>0} a_i w_i w_i^T = \sum_{a_i<0} (-a_i) w_i w_i^T.$$

Both sides are positive semidefinite. The range of a positive weighted sum of $w_i w_i^T$ is precisely the span of those w_i: its kernel consists of vectors perpendicular to every such w_i. Consequently the two spans coincide, and the affine hull H of all v_i in the circuit satisfies

$$1 + \dim H = \dim \mathrm{span}\{w_i : i \in I\} \leq \min(|\{a_i > 0\}|, |\{a_i < 0\}|) \leq 3.$$

This is an elementary positive-Gram range argument, not a new general circuit bound. Distinct points here cannot be proportional: equal ellipse radius forces the proportionality factor to be ± 1, and the positive first coordinate excludes minus one. Also an affine line meets this positive-definite ellipsoid in at most two points, by its strictly quadratic restriction to the line. If $\dim H \leq 1$, there are at most two points, whose rank-one matrices are independent by nonproportionality, a contradiction. It remains that $\dim H = 2$.

If $0 \in H$, this is a two-dimensional linear plane. Its rank-one matrices span a space of dimension at most three, so minimality now gives $s \leq 4$. For any observation b nonconstant on I, (4.3) and the total vector relation give nonzero-coefficient linear relations on both nonempty sets $I_1 = \{i : b(i) = 1\}$ and $I_0 = \{i : b(i) = 0\}$. Nonproportionality forces each set to contain at least three points. This would give $s \geq 6$, a contradiction. If instead every observation in $\mathcal{B}$ were constant on I, then minimality excludes $s = 1$ (a one-element circuit is impossible because $v_i v_i^T \neq 0$), so we may choose distinct $i, j \in I$; since $\mathcal{B}$ separates every pair of distinct indices of J, some $b \in \mathcal{B}$ satisfies $b(i) \neq b(j)$, contradicting constancy on I.

If $0 \notin H$, write $H = \{v : c \cdot v = 1\}$. Any three distinct points of H on the ellipsoid are linearly independent: a linear relation has coefficient sum zero after applying c, hence would make the three points affinely collinear, which the line-intersection fact forbids. A nonconstant observation would therefore require at least four points on each of I_1, I_0, yielding $s \geq 8$, again a contradiction. Thus every observation is constant on I. A one-element circuit is impossible because $v_i v_i^T \neq 0$; separation of any two indices in I is contradicted. □

Theorem 4.3 (Every actual positive clique has at most six types)**.** *For every actual maximizer, and every positive clique J of its full support, the map $a \mapsto \sum_{i \in J} a_i v_i v_i^T$ is injective. In particular $|J| \leq 6$. If $\{v_i : i \in J\}$ lies in a linear plane, then $|J| \leq 3$. If it lies in an affine plane not containing zero, then $|J| \leq 5$.*

Proof. Use the actual observations $b_k(i) = B_{ki}$, for all rows k. For nonempty J, a row $k \in J$ supplies the constant-one observation. They separate indices: B is symmetric and has distinct full rows. Lemma 4.2 now applies by (4.2). The linear-plane refinement follows from $\dim \mathrm{Sym}_2 = 3$. For an affine plane $c \cdot v = 1$, the nonzero symmetric matrix $L - \mathcal{M} c c^T$ is orthogonal to every $v_i v_i^T$. It is nonzero because L has rank three while $\mathcal{M} c c^T$ has rank at most one. This leaves dimension at most five. Empty cliques require no argument. □

Corollary 4.4 (Proof of Theorem 1.2)**.** *Every maximizer has at most 24 distinct canonical support points.*

Proof. Partition its full support by the signs of coordinates two and three, assigning a zero coordinate to the positive sign. There are at most four parts. Within each part, the last two coordinate products are nonnegative and the first-coordinate product is strictly positive. Each part is a positive clique. Theorem 4.3 gives $m \leq 4 \cdot 6 = 24$ on its original support. This proves Theorem 1.2. □

5 Moments and regression for one isotropic law

We now turn to the numerical estimate. Throughout this section the law is arbitrary and isotropic. Bounded positive radii will be imposed only when a radial comparison requires them. For later bookkeeping define

$$d_0 = -\frac{25624284539519}{53655078400000} + \frac{\sqrt{5}}{2} - \frac{41\sqrt{3}}{120}, \qquad d_1 = d_0 + \frac{103}{9000000}, \qquad d = d_1 + \frac{1}{3000000}. \tag{5.1}$$

Then $d = (3+\sqrt{5})/2 - 2c$, with c as in (1.2). The intermediate constants separate the joint moment estimate from the final atomic term. Fix $\mathbb{E}XX^T = I_3$, let $R = |X|$, and put $U = X/R$ when $R > 0$. At $R = 0$ choose any fixed unit vector for U. Define

$$C = \mathbb{E}R, \quad V = 3 - C^2, \quad \mathbf{m} = \mathbb{E}X, \quad B = |\mathbf{m}|^2, \quad e = 1 - B,$$
$$A = \mathbb{E}|X \cdot X'|, \qquad D = \frac{3+\sqrt{5}}{2} - A - B.$$

Here $V, e \geq 0$: $\mathbb{E}R^2 = 3$ and Cauchy–Schwarz gives $(v \cdot \mathbf{m})^2 \leq \mathbb{E}(v \cdot X)^2 = 1$ for every unit v. All pair expectations below are over independent copies of this same law. Set

$$M = \mathbb{E}[RUU^T] = \frac{C}{3} I_3 + Z, \qquad J = 1 - \operatorname{tr}(M^2) = \frac{V}{3} - \|Z\|_F^2.$$

The regression vector $\xi = U - MX$ satisfies

$$\mathbb{E}[X\xi^T] = 0, \quad \mathbb{E}\|\xi\|^2 = J \geq 0, \quad \mathbb{E}UU^T = M^2 + \mathbb{E}[\xi\xi^T]. \tag{5.2}$$

In particular J is an actual regression error, not a freely chosen moment.

Here are the harmonic normalizations used throughout. All tensor inner products sum over all ordered indices. Put

$$\Omega_{ijkl} = \frac{\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}}{15}, \qquad (\mathcal{L}_3(y))_{ijk} = \frac{y_i\delta_{jk} + y_j\delta_{ik} + y_k\delta_{ij}}{5}.$$

For symmetric traceless Z, put

$$(\mathcal{L}_4(Z))_{ijkl} = \frac{1}{7}(\delta_{ij}Z_{kl} + \delta_{ik}Z_{jl} + \delta_{il}Z_{jk} + \delta_{jk}Z_{il} + \delta_{jl}Z_{ik} + \delta_{kl}Z_{ij}).$$

For unit u, set

$$\Psi_3(u) = u^{\otimes 3} - \mathcal{L}_3(u), \qquad \Psi_4(u) = u^{\otimes 4} - \Omega - \mathcal{L}_4(uu^T - I_3/3),$$
$$T_i = \mathbb{E}[R\Psi_i(U)], \qquad H = \|T_3\|_F, \qquad K = \|T_4\|_F.$$

Contraction in two indices gives the orthogonal decompositions

$$\mathbb{E}[RU^{\otimes 3}] = T_3 + \mathcal{L}_3(\mathbf{m}), \qquad \mathbb{E}[RU^{\otimes 4}] = T_4 + C\Omega + \mathcal{L}_4(Z).$$

Indeed $\operatorname{ct}\Omega = I_3/3$, $\|\Omega\|^2 = 1/5$, $\operatorname{ct}\mathcal{L}_4(Z) = Z$, and $\langle T, \mathcal{L}_4(Z)\rangle = (6/7)\langle \operatorname{ct} T, Z\rangle$. Also $\operatorname{ct}\mathcal{L}_3(y) = y$ and $\langle N, \mathcal{L}_3(y)\rangle = (3/5)\operatorname{ct} N \cdot y$. It follows directly that

$$\begin{aligned} &\langle \Psi_3(u), \Psi_3(v)\rangle = g_3(u \cdot v), \qquad g_3(t) = t^3 - 3t/5, \qquad \|\Psi_3(v)\|^2 = 2/5, \\ &\langle \Psi_4(u), \Psi_4(v)\rangle = g_4(u \cdot v), \qquad g_4(t) = t^4 - 6t^2/7 + 3/35, \qquad \|\Psi_4(v)\|^2 = 8/35. \end{aligned} \tag{5.3}$$

The tensor moments involve R, not R^3 or R^4, and are integrable under $\mathbb{E}R^2 = 3$.

Write $s = \sqrt{5}$ and define the scalar absolute-value majorant and its slack by

$$a = \frac{1+s}{6}, \quad b = \frac{7s-5}{8}, \quad \rho = \frac{15s-25}{16}, \qquad h_0(z) = \frac{(1-z)(sz-1)^2(sz+s+2)}{2(s+1)^2}.$$

Expansion gives, for $|t| \le 1$,

$$a + b(t^2 - 1/3) - \rho(t^4 - 1/5) - |t| = h_0(|t|) \ge 0.$$

This majorant is from [12, Lemma 4.4]. Let $S = \mathbb{E}[RR' h_0(|U \cdot U'|)]$. The decompositions above give

$$A = aC^2 + b\|Z\|^2 - \rho\left(K^2 + \frac{6}{7}\|Z\|^2\right) - S.$$

Consequently

$$D = e + \alpha V + \beta J + \rho K^2 + S, \qquad \alpha = \frac{2s-1}{14}, \quad \beta = \frac{s+10}{14}. \tag{5.4}$$

Useful rational bounds are $\alpha \ge 26/105$, $\beta \ge 6/7$, and $\rho \ge 1/2$, using $s > 67/30$, $s \ge 2$, and $s \ge 11/5$.

5.1 A regression estimate

Suppose $B > 0$. Put

$$x = \sqrt{B}, \quad v = \mathbf{m}/x, \quad t = U \cdot v, \quad Y = Rt, \quad q_0 = 1/\sqrt{3}, \quad q = C/(3x), \quad L = \mathbb{E}[R(t-q_0)^2].$$

Then $\mathbb{E}Y = x$, $\mathbb{E}Y^2 = 1$, $\operatorname{Var} Y = e$. If $w_0 = \mathbb{E}\xi$, (5.2) gives $w_0 = \mathbb{E}[(1 - \mathbf{m} \cdot X)\xi]$, hence $|w_0|^2 \le eJ$. For traceless symmetric 3 by 3 matrices, $|Zv|^2 \le (2/3)\|Z\|^2$; the other two eigenvalues accompanying an eigenvalue z have squared sum at least $z^2/2$. Writing $J_v = \mathbb{E}(v \cdot \xi)^2 \le J$, we obtain

$$\begin{aligned} \mathbb{E}(t-q)^2 &= |Mv - q\mathbf{m}|^2 + q^2 e + J_v - 2qv \cdot w_0 \\ &\le 2V/9 + J/3 + q^2 e + 2q\sqrt{eJ} =: F_0, \\ \mathbb{E}(Y - qR)^2 &\le V/3 + 3q^2 e + 2q\sqrt{eV} =: G_0. \end{aligned}$$

Here $Mv - q\mathbf{m} = Zv$ and $\mathbb{E}RY \ge Cx - \sqrt{eV}$. Cauchy–Schwarz gives $L_q := \mathbb{E}[R(t-q)^2] \le \sqrt{F_0 G_0}$. Note that $Rt = R(U \cdot v) = (RU) \cdot v = X \cdot v = Y$, so $\mathbb{E}[Rt] = \mathbb{E}Y = x$; this is the only place at which the regression vector enters the exact identity below. With $\zeta = C/\sqrt{3}$,

$$L_q - L = q_0(\zeta - x)^2(\zeta + 2x)/x^2 \ge 0.$$

Thus, writing $u = \sqrt{e}$, $z = \sqrt{V}$, $j = \sqrt{J}$,

$$L \le \overline{L} := \frac{3}{4}F_0 + \frac{1}{3}G_0 = \frac{7q^2}{4}u^2 + \frac{5}{18}z^2 + \frac{1}{4}j^2 + \frac{2q}{3}uz + \frac{3q}{2}uj. \tag{5.5}$$

Set $r_{\mathrm{reg}} = \overline{L} - L \ge 0$. This remainder will be kept in the joint identity. No lower radius bound has been used here.

6 A signed quartic and a joint defect identity

Define

$$f(w) = \frac{8352 + 43806w + 54016w^2 - 12928w^3 - 10240w^4}{41503}, \qquad \sigma(w) = f(w) - |w| - w.$$

The polynomial is the degree-at-most-four Hermite interpolant with double contacts to $|w|+w$ at $-3/8$ and $1/2$, and a contact at 1. Uniqueness follows because a difference of two such polynomials would have five zeros counted with multiplicity. The following explicit identities prove the needed inequalities; interpolation alone is not a majorant proof.

Lemma 6.1. *For every $w \in [-1, 1]$,*

$$\sigma(w) \geq 0, \qquad \sigma(w) \geq \frac{1 - 9w^2}{25}. \tag{6.1}$$

Proof. Direct expansion gives

$$41503\sigma(w) = \begin{cases} (1-w)(2w-1)^2(8352 + 2560w), & 0 \leq w \leq 1, \\ (8w+3)^2(928 - 82w - 160w^2), & -1 \leq w \leq 0. \end{cases}$$

The final factor on the negative interval is concave and at least 850. This proves nonnegativity, including the join at 0 and both endpoints. The second assertion follows immediately when $|w| \geq 1/3$. For $w = -z$, $0 \leq z \leq 1/3$, write

$$\sigma(-z) = (z - 3/8)^2 \frac{59392 + 5248z - 10240z^2}{41503} \geq \frac{10}{7}(z - 3/8)^2.$$

Indeed $z(5248 - 10240z) \geq 0$ on this interval and $59392/41503 > 10/7$. The remaining comparison is the exact square

$$\frac{10}{7}(z - 3/8)^2 - \frac{1 - 9z^2}{25} = \frac{313}{175}\left(z - \frac{375}{1252}\right)^2 + \frac{109}{250400} > 0.$$

For $w = z$, $0 \leq z \leq 1/3$, the positive-interval factorization gives

$$\sigma(z) = (z - 1/2)^2 \frac{(1-z)(33408 + 10240z)}{41503} \geq \frac{1}{2}(z - 1/2)^2,$$

because the coefficient is at least $22272/41503 > 1/2$. Now use

$$\frac{1}{2}(z - 1/2)^2 - \frac{1 - 9z^2}{25} = \frac{43}{50}\left(z - \frac{25}{86}\right)^2 + \frac{53}{4300} > 0.$$

These intervals exhaust $[-1, 1]$. □

Put $P = \mathbb{E}[RR'\sigma(U \cdot U')]$ for the same actual law as before, and set

$$a_3 = \frac{12928}{41503}, \quad a_4 = \frac{10240}{41503}, \quad b_e = \frac{16386}{18865},$$
$$b_V = \frac{64608}{290521}, \quad b_J = \frac{316672}{290521}, \quad \delta_{\mathrm{q}} = \frac{3 + \sqrt{5}}{2} - \frac{544886}{207515}.$$

Integration of the expanded f, using (5.3), gives the single exact joint budget

$$D = \delta_{\mathrm{q}} + b_e e + b_V V + b_J J + a_3 H^2 + a_4 K^2 + P. \tag{6.2}$$

For example, writing $f = c_0 + c_1 w + c_2 w^2 - a_3 w^3 - a_4 w^4$, the coefficients are $b_e = c_1 - 3a_3/5$, $b_V = c_0 + 3a_4/35$, $b_J = c_2 - 6a_4/7$, and the integrated constant is $3(c_0 - a_4/5) + b_e + c_2 = 544886/207515$. All tensors are R-weighted and σ is bounded, so this equality uses no radial moment beyond the second one. Identity (5.4) will only justify a small-D branch; it is not added as an independent resource to (6.2).

Lemma 6.2. *Suppose $0 < r_- \le R \le r_+ < \infty$ almost surely and $r_+/r_- \le 3$. Then*

$$P = \frac{3}{50}(V - 9J) + r_{\mathrm{rad}}, \qquad r_{\mathrm{rad}} \ge 0. \tag{6.3}$$

Proof. For $w = U \cdot U'$, the exact pair identities from the full Gram are

$$V = \frac{1}{2}\mathbb{E}(R - R')^2, \qquad J = \frac{1}{2}\mathbb{E}[(R - R')^2 w^2].$$

The latter uses $\mathbb{E}[R^2 w^2] = \mathbb{E}[(R')^2 w^2] = 1$ and $\mathbb{E}[RR' w^2] = \mathrm{tr}(M^2)$; it assumes no independence of radius and direction. The ratio bound gives $(R - R')^2 \le (4/3)RR'$. Use the two conclusions of Lemma 6.1 in this order:

$$P \ge \frac{3}{4}\mathbb{E}[(R - R')^2 \sigma(w)] \ge \frac{3}{100}\mathbb{E}[(R - R')^2(1 - 9w^2)] = \frac{3}{50}(V - 9J).$$

The expression $V - 9J$ need not be nonnegative. Define r_{rad} as the actual difference; only this difference is claimed nonnegative. □

7 Directional and radial control

Throughout this section assume $0 < r_- \le R \le r_+ < \infty$ almost surely, with $r_+/r_- \le 3$, and work in the branch $D < 1/20$. Equation (5.4) implies

$$e < 1/20, \quad x > 9/10, \quad V < 21/104 < 2/9, \quad C > 5/3, \qquad q^2 = \frac{3 - V}{9(1 - e)} \le \frac{20}{57}. \tag{7.1}$$

In particular $\mathbf{m} \neq 0$. Put $p_i = \langle T_i, \Psi_i(v) \rangle = \mathbb{E}[R g_i(t)]$. The elementary Hermite identities

$$\begin{aligned} \left(\frac{53}{28}t^2 - \frac{9}{4}t + \frac{41}{70}\right) - g_4(t) &= (1 - t)(t - 1/2)^2(t + 2) \ge 0, \\ (2t^2 - 37t/20 + 1/4) - g_3(t) &= (1 - t)(t - 1/2)^2 \ge 0 \end{aligned} \tag{7.2}$$

hold on $[-1, 1]$. Define

$$\gamma_4 = 9/4 - 53\sqrt{3}/42, \quad \ell_4 = 9/4 - 511\sqrt{3}/420, \quad \ell_3 = 37/20 - 11\sqrt{3}/12.$$

Here $0 < \gamma_4 < 1/15$ (square the positive rational comparisons). Since $\mathbb{E}Rt = x$, $\mathbb{E}Rt^2 = L + 2x/\sqrt{3} - C/3$, the first identity gives

$$-p_4 \ge \gamma_4 x + 19C/420 - 53L/28 \ge \ell_4 - e/15 - 19V/1400 - 53L/28.$$

We used $x \ge 1 - e$ and $C \ge \sqrt{3} - 3V/10$; the latter follows from $V = (\sqrt{3} - C)(\sqrt{3} + C)$ and $C > 5/3$. The second identity gives

$$-p_3 \ge (37/20 - 4/\sqrt{3})x + 5C/12 - 2L \ge \ell_3 + 9e/40 - V/8 - 2L.$$

Here $4/\sqrt{3} - 37/20 > 9/20$ and $x \le 1 - e/2$. Define the nonnegative differences in these last two inequalities by

$$\begin{aligned} r_4 &= -p_4 - \ell_4 + e/15 + 19V/1400 + 53L/28, \\ r_3 &= -p_3 - \ell_3 - 9e/40 + V/8 + 2L. \end{aligned}$$

The polynomial identities, rather than a fixed sphere assumption, justify these directional estimates for the actual weighted moments.

Set $\eta = 1/10$, $\theta = 6/25$, and $\ell = 53\eta/28 + 2\theta = 937/1400$. Define the tensor squares

$$\mathcal{R}_4 = a_4 \left\| T_4 + \frac{\eta}{2a_4}\Psi_4(v) \right\|^2, \qquad \mathcal{R}_3 = a_3 \left\| T_3 + \frac{\theta}{2a_3}\Psi_3(v) \right\|^2.$$

Their exact expansions give

$$a_4 K^2 = -\eta p_4 - \frac{2\eta^2}{35a_4} + \mathcal{R}_4, \qquad a_3 H^2 = -\theta p_3 - \frac{\theta^2}{10a_3} + \mathcal{R}_3.$$

Define

$$E = \frac{5183683}{5659500}, \quad V_c = \frac{145858703}{581042000}, \quad J_c = \frac{7989533}{14526050}, \quad \mathcal{R}_0 = \mathcal{R}_3 + \mathcal{R}_4 + \eta r_4 + \theta r_3 \geq 0.$$

Substituting the directional identities and (6.3) into (6.2), then writing $L = \overline{L} - r_{\mathrm{reg}}$, gives the exact identity

$$D = d_0 + Ee + V_c V + J_c J - \ell\overline{L} + \mathcal{R}_0 + r_{\mathrm{rad}} + \ell r_{\mathrm{reg}}. \tag{7.3}$$

Here

$$d_0 = \delta_{\mathrm{q}} + \eta\ell_4 + \theta\ell_3 - \frac{2\eta^2}{35a_4} - \frac{\theta^2}{10a_3},$$

which simplifies to (5.1). The coefficient checks are

$$E = b_e - \eta/15 + 9\theta/40, \quad V_c = b_V - 19\eta/1400 - \theta/8 + 3/50, \quad J_c = b_J - 27/50.$$

All the remainders in this identity refer to the same probability law.

Put $B_c = V_c - 5\ell/18$ and $C_c = J_c - \ell/4$. The exact rational values are

$$B_c = \frac{42565069}{653672250}, \qquad C_c = \frac{88944217}{232416800}.$$

For $u = \sqrt{e}$, $z = \sqrt{V}$, and $j = \sqrt{J}$, the remaining quadratic has the explicit completion

$$\begin{aligned} Ee + V_c V + J_c J - \ell\overline{L} = {} & \frac{13}{200}\left(z - \frac{937}{273}qu\right)^2 + \frac{19}{50}\left(j - \frac{2811}{2128}qu\right)^2 \\ & + \left(\frac{183}{200} - \frac{725029049}{278853120}q^2\right)e \\ & + \frac{10481}{11319000}e + \frac{305491}{2614689000}V + \frac{625833}{232416800}J. \end{aligned} \tag{7.4}$$

Indeed the last three fractions are $E - 183/200$, $B_c - 13/200$, and $C_c - 19/50$. Also

$$\frac{725029049}{278853120} = \frac{7\ell}{4} + \frac{\ell^2}{9(13/200)} + \frac{9\ell^2}{16(19/50)}.$$

By (7.1), the coefficient of e in parentheses is at least

$$\frac{183}{200} - \frac{20}{57}\frac{725029049}{278853120} = \frac{53754367}{19868284800} > 0.$$

Thus every term on the right of (7.3) after d_0 is nonnegative. The displayed squares give an explicit certificate for the remaining quadratic form.

Proposition 7.1. *Every isotropic law on $\mathbb{R}^3$ whose radii lie almost surely in a positive interval of endpoint ratio at most 3 satisfies $D \geq d_0$. In the branch $D < 1/20$, (7.3)–(7.4) give the full common-remainder certificate.*

Proof. The small branch was proved above. Otherwise $D \geq 1/20 > d_0$. For explicit exact comparisons use $682/305 < \sqrt{5} < 161/72$ and $433/250 < \sqrt{3} < 97/56$, each verified by squaring. They yield

$$d_0 - 7/144 > \frac{940735364069}{29456638041600000} > 0, \qquad 1/20 - d_0 > \frac{620701786231}{482895705600000} > 0.$$

No $B = 0$ division occurs: it is excluded by $e < 1/20$ in the small branch. □

8 A reserve in the quartic slack

Throughout this section assume $0 < r_- \le R \le r_+ < \infty$ almost surely, with $r_+/r_- \le \sqrt{8}$. Write $g(w) = (1 - 9w^2)/25$. Then

$$\frac{(R-R')^2}{RR'} \le \sqrt{8} + \frac{1}{\sqrt{8}} - 2 = \frac{9\sqrt{2}}{4} - 2 \le \frac{32}{27}.$$

For the last inequality, $\sqrt{2} < 344/243$ follows by squaring, with $(344/243)^2 - 2 = 238/59049 > 0$. Put $z_{\mathrm{rad}} = 3(R-R')^2/(4RR')$, so $0 \le z_{\mathrm{rad}} \le 8/9$. The actual remainder in (6.3) now has the exact decomposition

$$r_{\mathrm{rad}} = \frac{P}{9} + r_{\mathrm{tr}}, \qquad r_{\mathrm{tr}} = \mathbb{E}\big[RR'\,\{(8/9 - z_{\mathrm{rad}})\sigma(w) + z_{\mathrm{rad}}(\sigma(w) - g(w))\}\big] \ge 0. \tag{8.1}$$

This uses both pointwise inequalities of Lemma 6.1, including where $g < 0$. It is not a second independent budget or a new copy of P.

8.1 A contact polynomial without a cubic term

Using the same interpolation nodes and tensor weights, set

$$h_*(t) = (t-1)(t-1/2)(t+3/8), \qquad z_*(t) = (t+9/8)h_*(t) = t^4 - \frac{85}{64}t^2 + \frac{15}{128}t + \frac{27}{128}.$$

The factor $t + 9/8$ cancels the cubic term. We need the following elementary bound, for which we give an explicit proof.

Lemma 8.1. *For $-1 \le t \le 1$, $z_*(t)^2 \le 3\sigma(t)/2$.*

Proof. For $0 \le t \le 1$, cancel common factors away from their zeros to obtain

$$\frac{z_*(t)^2}{\sigma(t)} = \frac{41503}{4(8352 + 2560t)} q_*(t), \qquad q_*(t) = (1-t)(t+3/8)^2(t+9/8)^2.$$

The cancelled expressions extend continuously at the zeros. Its derivative is

$$q_*'(t) = (t+3/8)(t+9/8)(165/64 - t/2 - 5t^2).$$

The last factor decreases on $[0,1]$, equals $13/576 > 0$ at $2/3$, and equals $-33/256 < 0$ at $11/16$. Thus the maximum is within $[2/3, 11/16]$. On that interval $(t+3/8)(t+9/8) < 2$ and $q_*'(t) \le 13/288$. The interval has length $1/48$, so

$$q_*(t) \le \frac{1155625}{995328} + \frac{13}{13824} < \frac{6}{5};$$

the final positive difference is $189163/4976640$. Consequently $z_*^2/\sigma \le 41503/27840 < 3/2$ on this half interval. For $t = -x$, $0 \le x \le 1$, the other factorization gives

$$\frac{z_*(-x)^2}{\sigma(-x)} = \frac{41503\, p_*(x)^2}{64(928 + 82x - 160x^2)}, \qquad p_*(x) = (1+x)(x+1/2)(9/8 - x) \ge 0.$$

The quadratic denominator factor is at least 850, by concavity and its endpoint values. With $y = x - 1/2 \in [-1/2, 1/2]$,

$$p_*(x) = 15/16 + y/16 - 15y^2/8 - y^3 \le 15/16 + y/16 - 11y^2/8 = \frac{1321}{1408} - \frac{11}{8}(y - 1/44)^2 < 1.$$

It follows that $z_*^2/\sigma \le 41503/54400 < 3/2$ here. At every cancelled zero the claimed inequality follows directly or by continuity. These two intervals cover the domain. □

Remark 8.2. The constant $3/2$ is not sharp: a dense grid scan gives $\sup_{[0,1]} z_*^2/\sigma = 1.200224\ldots$ and $\sup_{[-1,0]} z_*^2/\sigma = 0.615384\ldots$. We keep $3/2$ because it keeps every subsequent coefficient rational and hand-verifiable. The loss enters the certificate only through r_{ct} and through the divisor 125 in (8.4); both are quantified in Appendix A.

In particular the following displayed square proves a useful linear payment:

$$\sigma(t) - \frac{z_*(t)}{125} + \frac{3}{125000} = \left(\sigma(t) - \frac{2}{3}z_*(t)^2\right) + \frac{2}{3}\left(z_*(t) - \frac{3}{500}\right)^2 \geq 0. \tag{8.2}$$

All these functions are bounded on $[-1,1]$. Integrating against the actual nonnegative weight RR' uses only $\mathbb{E}R < \infty$, not a fourth or eighth radial moment. The actual contact moment is

$$Z_* := \mathbb{E}[RR' z_*(U \cdot U')] = K^2 + \frac{7}{320} - \frac{15}{128}e - \frac{561}{4480}V + \frac{211}{448}J. \tag{8.3}$$

Indeed the RR' moments of $1, w, w^2, w^4$ are respectively $3 - V, 1 - e, 1 - J, K^2 + 6\|Z\|^2/7 + (3 - V)/5$, all for this same law. Define

$$\mathcal{C} = Z_*/125 - 3C^2/125000, \qquad r_{\mathrm{ct}} = P - \mathcal{C} \geq 0. \tag{8.4}$$

The scalar $\mathcal{C}$ need not itself be nonnegative. Its exact contribution is

$$\frac{\mathcal{C}}{9} = \frac{103}{9000000} + \frac{K^2}{1125} - \frac{e}{9600} + \left(-\frac{187}{1680000} + \frac{1}{375000}\right)V + \frac{211}{504000}J. \tag{8.5}$$

8.2 The full joint certificate

For $D < 1/20$ define the explicitly nonnegative quadratic expression

$$\begin{aligned} Q = {} & \frac{13}{200}\left(\sqrt{V} - \frac{937}{273}q\sqrt{e}\right)^2 + \frac{19}{50}\left(\sqrt{J} - \frac{2811}{2128}q\sqrt{e}\right)^2 \\ & + \left(\frac{183}{200} - \frac{725029049}{278853120}q^2\right)e \\ & + \frac{148831}{181104000}e + \frac{42847433}{5229378000000}V + \frac{65082103}{20917512000}J. \end{aligned} \tag{8.6}$$

The parenthetical coefficient is positive by (7.1) and the same exact lower bound following (7.4). The last coefficients are obtained from those in (7.4), respectively by adding $-1/9600$, $-187/1680000 + 1/375000$, and $211/504000$. Substitution of (8.1)–(8.5) into (7.3) yields the exact identity

$$D = d_1 + Q + \mathcal{R}_0 + \ell r_{\mathrm{reg}} + \frac{K^2}{1125} + r_{\mathrm{tr}} + \frac{r_{\mathrm{ct}}}{9}. \tag{8.7}$$

Every remainder is defined from the same X, its independent copy and its complete Gram matrix.

Proposition 8.3. *Every isotropic law whose radii lie almost surely in a positive bounded interval of endpoint ratio at most $\sqrt{8}$ satisfies $D \geq d_1$. In the branch $D < 1/20$, (8.7) retains all displayed remainders.*

Proof. In the small branch all terms after d_1 are nonnegative. In the other branch $D \geq 1/20 > d_1$. To verify the last comparison entirely by rationals, use $\sqrt{5} < 161/72$ and $\sqrt{3} > 433/250$ in (5.1); they give

$$\frac{1}{20} - d_1 > \frac{1025292188593}{804826176000000} > 0.$$

No division by $B = 0$ occurs in the small branch, and the other branch requires none. The radial hypothesis will be verified for a maximizing law in Section 9, rather than imposed on every isotropic law. □

9 The coefficient bound and its transfer

We return to a maximizing graphon. Put

$$D_* = (3+\sqrt{5})/2 - 2\mathcal{M}.$$

By the threshold identity in Lemma 2.5, its canonical law $X = F$ satisfies $A + B = 2\mathcal{M}$, so the deficit D of the preceding sections equals D_*.

Lemma 9.1. *If W is an actual maximizer and $D_* = (3+\sqrt{5})/2 - 2\mathcal{M} < 1/20$, then $\lambda_3 > 44/375$, $\lambda_1 < 15/16$, and the canonical frame has positive bounded radii of endpoint ratio less than $\sqrt{8}$.*

Proof. The small branch and $\sqrt{5} > 682/305 > 559/250$ give

$$\mathcal{M} > \frac{3+\sqrt{5}}{4} - \frac{1}{40} > \frac{321}{250}.$$

Lemma 2.1 gives $\lambda_3 = \mathcal{M} - \Sigma_2(W) > 321/250 - 7/6 = 44/375$. For $a = \lambda_1$, the Hilbert–Schmidt trace and the constant-function trial give

$$\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \le \iint W^2 \le \iint W \le \lambda_1, \qquad \mathcal{M} \le a + \sqrt{2a(1-a)}, \quad 0 < a \le 1.$$

The function on the right is decreasing on $[15/16, 1]$; differentiation on the interior reduces this to $(2a-1)^2 > 2a(1-a)$ there. Moreover

$$\frac{15}{16} + \sqrt{\frac{15}{128}} < \frac{321}{250}, \qquad \left(\frac{321}{250} - \frac{15}{16}\right)^2 - \frac{15}{128} = \frac{11499}{4000000} > 0.$$

Thus $a < 15/16$. By the actual ellipsoid identity,

$$\sqrt{\mathcal{M}/\lambda_1} \le R \le \sqrt{\mathcal{M}/\lambda_3}, \qquad \frac{\operatorname{ess\,sup} R}{\operatorname{ess\,inf} R} \le \sqrt{\lambda_1/\lambda_3} < \sqrt{5625/704} < \sqrt{8},$$

where $8 - 5625/704 = 7/704$. No simplicity assumption was made on the positive eigenvalues; the argument uses the whole orthonormal frame. □

Remark 9.2. Lemma 6.2 is stated with endpoint ratio 3, whereas the present lemma supplies $\sqrt{8} < 3$ for an actual maximizer; the two hypotheses are therefore compatible, with slack. Using $\sqrt{8}$ in Lemma 6.2 would replace the factor $4/3$ in (6.3) by $32/27$, a 12.5% gain on P, which is not needed here.

9.1 Positive-mass self-pairs

By Theorem 1.2, write the canonical law as $\sum_{i=1}^{m} p_i \delta_{v_i}$ with $m \le 24$ and all $p_i > 0$. Set $R_i = |v_i|$, $u_i = v_i/R_i$, $w_{ij} = u_i \cdot u_j$, and

$$\kappa = \frac{3}{125000}, \qquad h(t) = \sigma(t) - \frac{z_*(t)}{125} + \kappa, \qquad \chi = \sum_i p_i^2 R_i^2.$$

Equation (8.2) gives $h \ge 0$ and $h(1) = \kappa$. The same contact remainder in (8.4) therefore has the exact split

$$r_{\mathrm{ct}} = \kappa\chi + r_{\mathrm{off}}, \qquad r_{\mathrm{off}} = \sum_{i \ne j} p_i p_j R_i R_j h(w_{ij}) \ge 0. \tag{9.1}$$

These self-pairs have positive product mass. They are not values assigned to the null diagonal $x = y$ of the underlying nonatomic space. The estimate $\chi \geq C^2/m$ is the usual Cauchy–Schwarz cardinality bound. For related weighted-energy arguments, see [1, Theorem 2.1 and the subsequent discussion].

In the branch $D_* < 1/20$, let

$$\widehat{Q} = Q - \frac{V}{9000000}.$$

It is nonnegative: in (8.6) only the last separate V coefficient changes, to

$$\frac{42847433}{5229378000000} - \frac{1}{9000000} = \frac{14088797}{1743126000000} > 0.$$

Proposition 9.3. *For every maximizing graphon with $D_* < 1/20$,*

$$\begin{aligned} D_* = d + \widehat{Q} + \mathcal{R}_0 + \ell r_{\mathrm{reg}} + \frac{K^2}{1125} + r_{\mathrm{tr}} \\ + \frac{\kappa}{9}\left(\chi - \frac{C^2}{24}\right) + \frac{r_{\mathrm{off}}}{9}. \end{aligned} \tag{9.2}$$

Every term after d is nonnegative. Consequently $\mathcal{M} \leq c$.

Proof. Lemma 9.1 verifies the radial hypothesis of Proposition 8.3. Substitute (9.1) once into (8.7). Since $C^2 = 3 - V$,

$$\frac{\kappa\chi}{9} = \frac{1}{3000000} - \frac{V}{9000000} + \frac{\kappa}{9}\left(\chi - \frac{C^2}{24}\right).$$

This gives (9.2). Moreover

$$\chi - \frac{C^2}{24} = \sum_i \left(p_i R_i - \frac{C}{m}\right)^2 + C^2\left(\frac{1}{m} - \frac{1}{24}\right) \geq 0.$$

No lower bound on an individual mass is needed. The complementary branch also gives $D_* \geq d$, since the rational estimates used above imply

$$\frac{1}{20} - d > \frac{1025023913201}{804826176000000} > 0.$$

Thus $D_* \geq d$ in both branches, or equivalently $\mathcal{M} \leq c$. □

The atomic term changes the coefficient by $1/6000000$ relative to $(3+\sqrt{5})/4 - d_1/2$. Most of the improvement in Theorem 1.1 comes from the preceding joint moment estimate, not from this last term.

9.2 All isotropic laws and the exact finite shift

Proof of Theorem 1.1. Any Borel law on $\mathbb{R}^3$ can be realized on a standard nonatomic probability space, by adjoining an independent uniform variable if necessary. For its isotropic X, the coordinates are an orthonormal three-frame and $W_X = \mathbf{1}_{\{X \cdot X' > 0\}}$ is a measurable symmetric binary kernel. Therefore

$$\frac{\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2}{2} = \iint W_X X \cdot X' \leq \Sigma_3(W_X) \leq \mathcal{M} \leq c.$$

This proves (1.4), including zero vectors and unbounded radii. The radial restriction was used only for a maximizing canonical law.

For a finite graph, the exact-rank Ky Fan formula reads

$$\Lambda_3(G) = \max_{Q=Q^T=Q^2,\ \operatorname{tr} Q=3} \operatorname{tr}(A_G Q).$$

Factor $Q = YY^T$ with orthonormal columns, let y_i be its rows, and let X be uniform on $\sqrt{n}\,y_1, \dots, \sqrt{n}\,y_n$. Then

$$\mathbb{E}XX^T = I_3, \qquad \mathbb{E}|X \cdot X'| = \|Q\|_1/n, \qquad |\mathbb{E}X|^2 = \mathbf{1}^T Q \mathbf{1}/n,$$

where $\|Q\|_1$ denotes the entrywise absolute sum. Since $Q_{ii} \geq 0$ and $\sum_i Q_{ii} = 3$, the zero adjacency diagonal gives

$$\operatorname{tr}(A_G Q) \leq \sum_{i \neq j} (Q_{ij})_+ = \frac{\|Q\|_1 + \mathbf{1}^T Q \mathbf{1}}{2} - 3 \leq cn - 3.$$

Taking the maximum proves (1.3). No positivity assumption on $\lambda_3(G)$ is used; repeated eigenvalues, zero rows and $n = 3$ cause no exception. □

10 Further questions

The explicit construction in Lemma 2.2 leaves the interval

$$\frac{9 + \sqrt{5} + 2\sqrt{35}}{18} \leq \mathcal{M} \leq 1.284673255723918505\ldots.$$

The lower endpoint comes from a weighted wheel, but its global optimality is not established here. The support bound reduces the structure of every maximizer to finitely many positive-mass types without classifying the possible templates. It does not imply that the binary matrix B is nonsingular, that every maximizer has six types, or that individual masses are uniformly bounded away from zero. Determining the optimum remains a separate problem.

The exact identity (9.2) also distinguishes a quantitative certificate from a qualitative equality obstruction. Any further decrease obtained from this certificate must bound a remaining term from below under the same constraints, without counting the same quartic remainder twice. The present argument does not use an unquantified stability gap.

A Constants: provenance and machine verification

A reader cannot tell from the statements alone which of the many integer coefficients in this paper are forced by an interpolation problem and which were chosen and then verified. We record the distinction.

Uniquely determined. (i) The polynomial f of §6 is the unique polynomial of degree at most four with double contact to $|w| + w$ at $w = -3/8$ and at $w = 1/2$ and a contact at $w = 1$; five conditions determine five coefficients, so once the nodes are fixed the integers in f (including $41503 = 7^3 \cdot 11^2$) are forced. (ii) z_* of §8 is determined by the same nodes together with the requirement that the cubic term vanish, i.e. by the factor $(t + 9/8)$. (iii) c_{w} of Lemma 2.2 is the value of an explicit one-variable function at the critical point $b = (7+5\sqrt{7})/126$. (iv) The four rational certificates of §7 and §9 are obtained by substituting the stated rational bounds for $\sqrt{5}$ and $\sqrt{3}$ into d_0, d_1, d; no search is involved.

Chosen, then verified exactly. (i) The nodes $-3/8$, $1/2$, 1 themselves, and the majorant h_0 taken from [12, Lemma 4.4]. (ii) The parameters $\ell = 937/1400$, $\eta = 1/10$,

$\theta = 6/25$ of §7 and the square coefficients of (7.4) and (8.6); these were selected so that all residual coefficients remain positive, and every positivity claim is then checked by exact rational arithmetic. (iii) The divisor 125 and $\kappa = 3/125000$ of (8.4), and the constant $3/2$ of Lemma 8.1; the quantified loss is discussed in Remark 8.2.

Verification protocol. Every displayed identity, factorization, completion of squares and "> 0" assertion was re-checked in exact arithmetic: rationals as exact fractions and $\sqrt{3}, \sqrt{5}$ in the quadratic fields $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$, never in floating point. Pointwise inequalities were additionally scanned on a $2 \times 10^6 + 1$ point grid, and the finite-graph transfer was checked on all 33864 graphs of order $3 \le n \le 6$. The verification scripts are listed in the accompanying audit report of 2 November 2026.

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